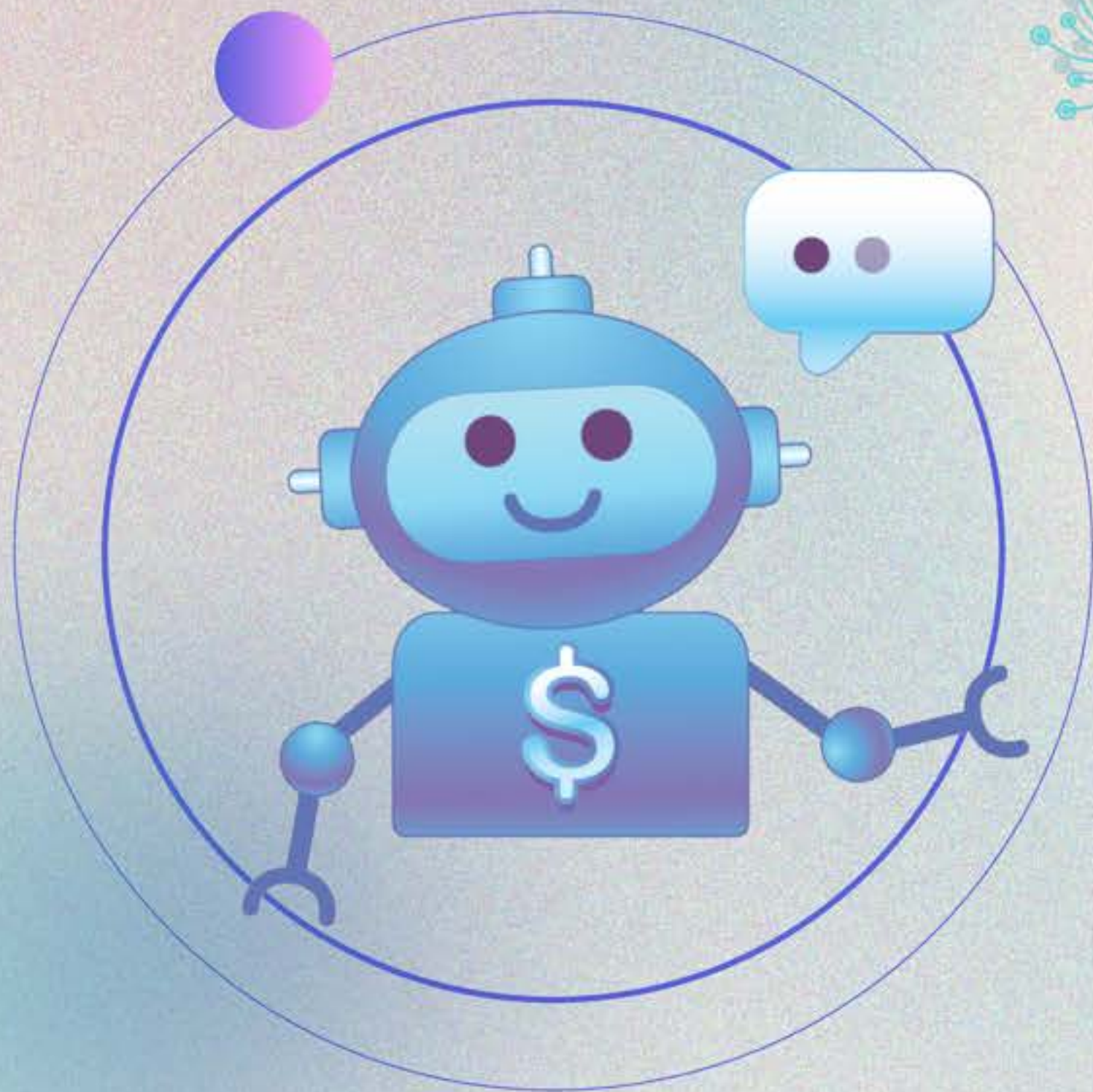




UMDACC DATATHON



404 FOUND



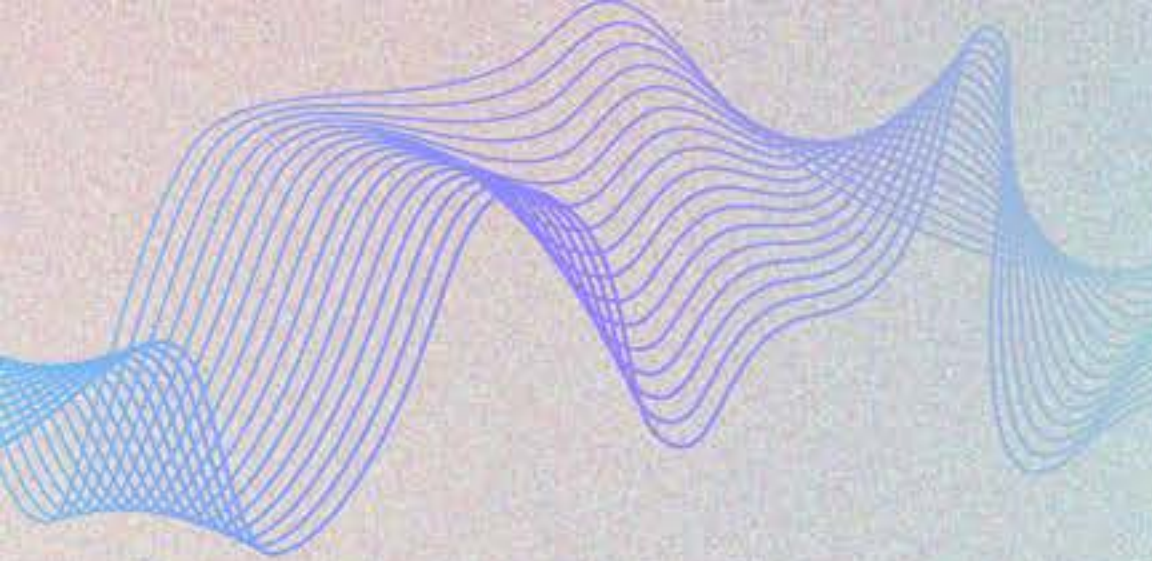
Kueh Pang Lang
Data Science Student
Universiti Malaya



Liew Jin Sze
Data Science Student
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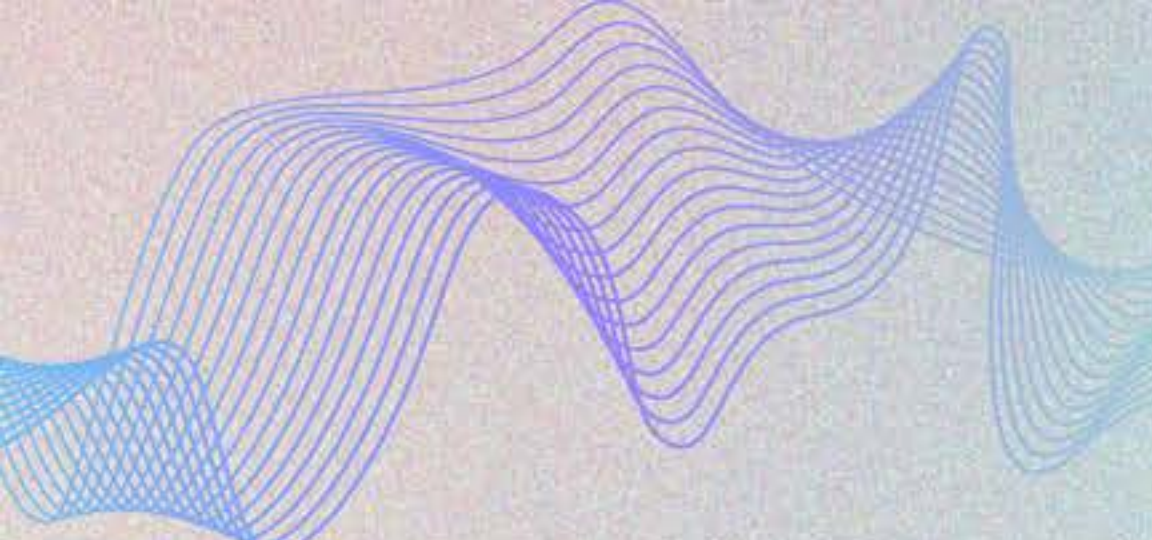
Kueh Pang Teng
Artificial Intelligence Student
Multimedia University



OUR MAIN FOCUS

Cryptocurrency Price Analysis

To understand and model the **key factors and anomalies** influencing cryptocurrency market behavior, enabling **better forecasting, anomaly detection, and insights** into market stability and manipulation.



ASSUMPTIONS

1. The dataset is accurate, reliable, and sufficient for answering the research questions.
2. While markets can be non-stationary, it is assumed that meaningful patterns and relationships exist over the studied time frame.
3. External events such as regulatory changes, macroeconomic shocks are assumed not to confound the results.
4. Data preprocessing steps (e.g., imputation, scaling) do not distort the relationships between variables.



OUR RESEARCH QUESTION

- Which factors are the most significant in predicting **hourly price movements** in the cryptocurrency market?
- How do these factors behave differently during periods of **high versus low market volatility**?
- Are there any **indicators** in the dataset that **signal potential market manipulation**, such as sudden spikes in exchange_whale_ratio?



DATA DESCRIPTION

1. Data Preprocessing

- data loading
- data information such as datatypes, columns and head
- identify categorical columns and numerical columns

2. Handling missing data

- check for missing data:
- drop columns '*fees_transaction_mean*' and '*fees_transaction_mean_usd*' as they contain more than 50% missing value
- drop rows with missing values
- apply KNN imputer on numeric columns

3. Data Transformation

- drop non-numeric columns *time* and *close_price* (target variable)
- apply standard scaling
- add back the target values *time* and *close_price*

1. Which factors are the most significant in predicting hourly price movements in the cryptocurrency market?



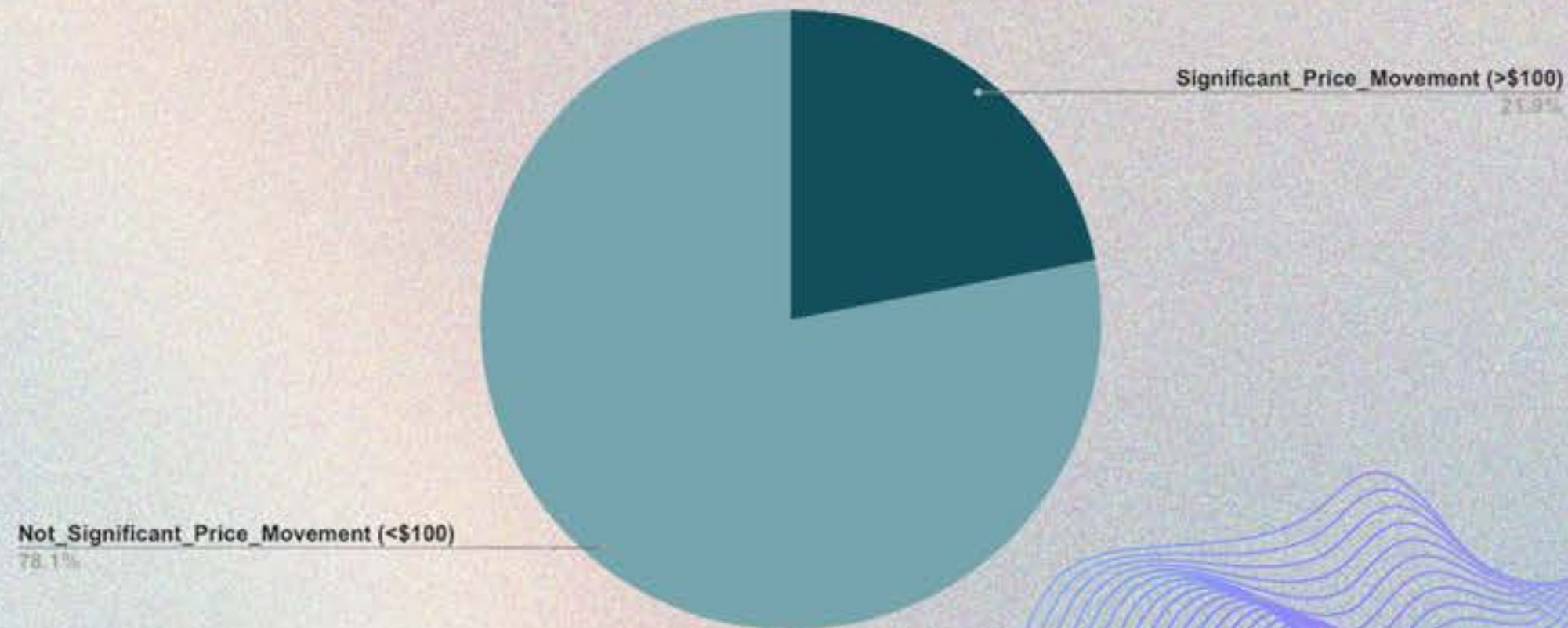
Price Movement indicate by
 $\text{price_change} = \text{close_price} - \text{open_price}$

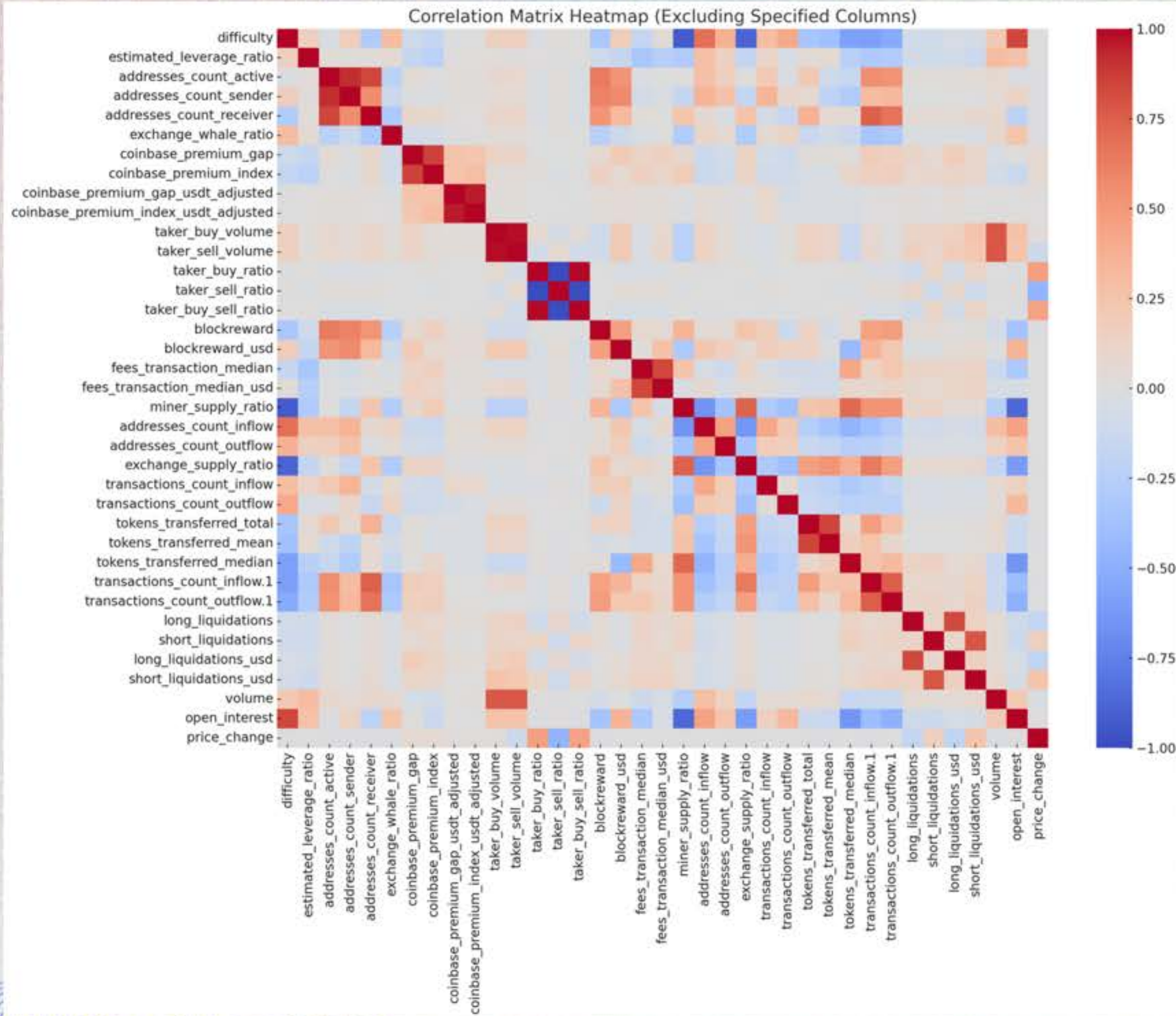
Methodology:

In this research, to find out which factors are the most significant in predicting hourly price movement, we define the **price movement indicate by price change in an hour**. To get further information and the concept, we do some EDA by conducting a **probability analysis** to reveal that the probability of significant price movement occur. We also **analyze correlation of variables** to identify potential predictors of price movement by calculating correlation coefficient between each variable and the target variable (price_change) using **SPSS**. Next, we develop a **linear regression model** using the selected variables as predictors and price_change as the dependent variable. Lastly, we identify the three most significant predictors based on **beta coefficients and p-values**.

Probability Analysis

- Based on our analysis, the probability of a significant price movement ($> \$100$) in an hour is **21.9%**.





Color Representation:

- The heatmap uses a gradient from blue to red, where:
 - Dark red : strong positive correlation (values close to +1).
 - Dark blue : strong negative correlation (values close to -1).
 - Light colors (white or pale) : weak or no correlation (values close to 0).

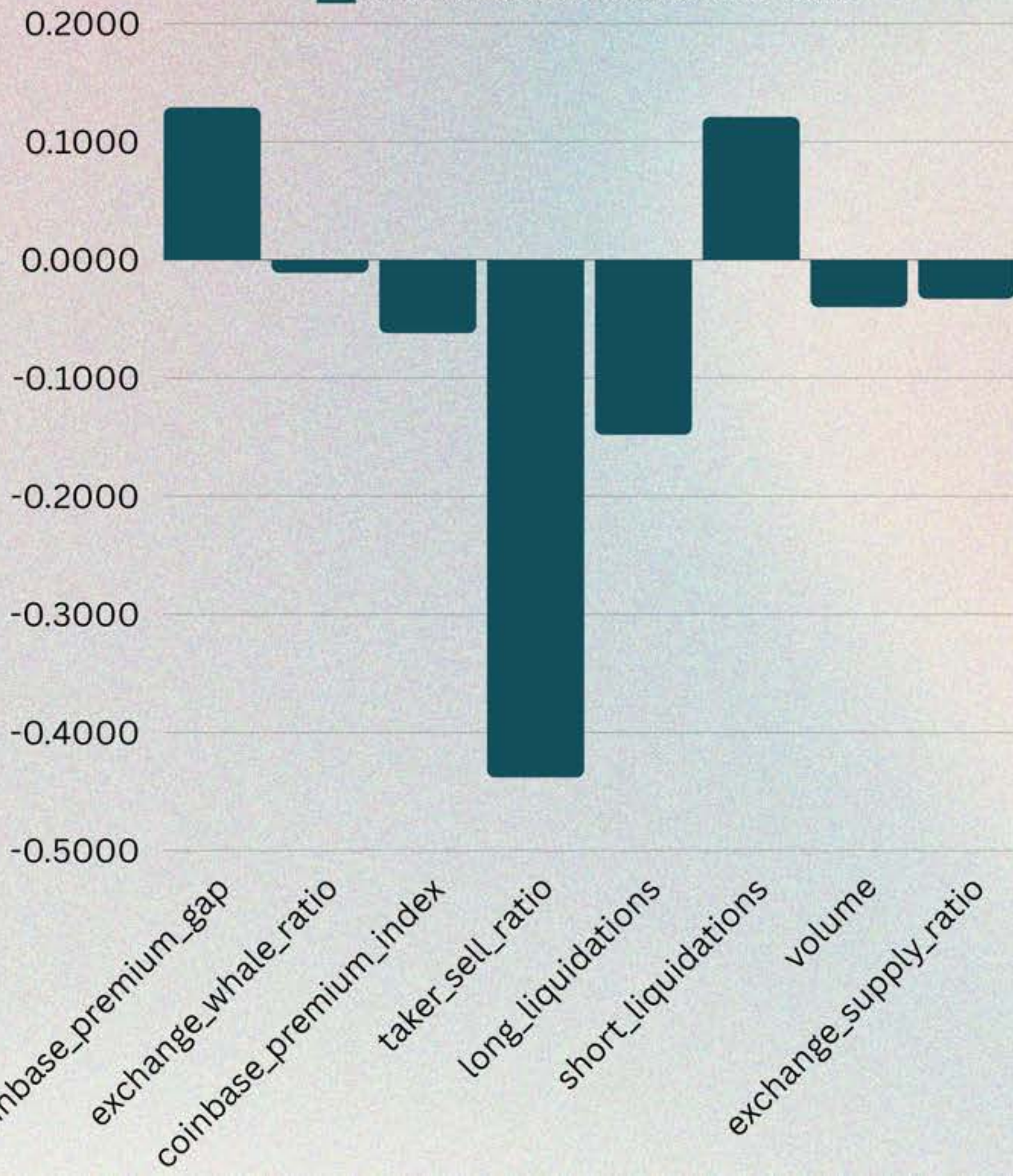
Key Observations:

We observed **11 variables** that show **significant correlation with the price_change variable**.

These variables will be used in the subsequent linear regression model to analyze their impact.

- coinbase_premium_gap
- exchange_whale_ratio
- coinbase_premium_index
- taker_sell_ratio
- long_liquidations
- short_liquidations
- volume
- addresses_count_active
- exchange_supply_ratio
- tokens_transferred_total
- transactions_count_inflow.1

Standardized Coefficients Beta



From the linear regression model:

- **Three Most Significant Variables:**

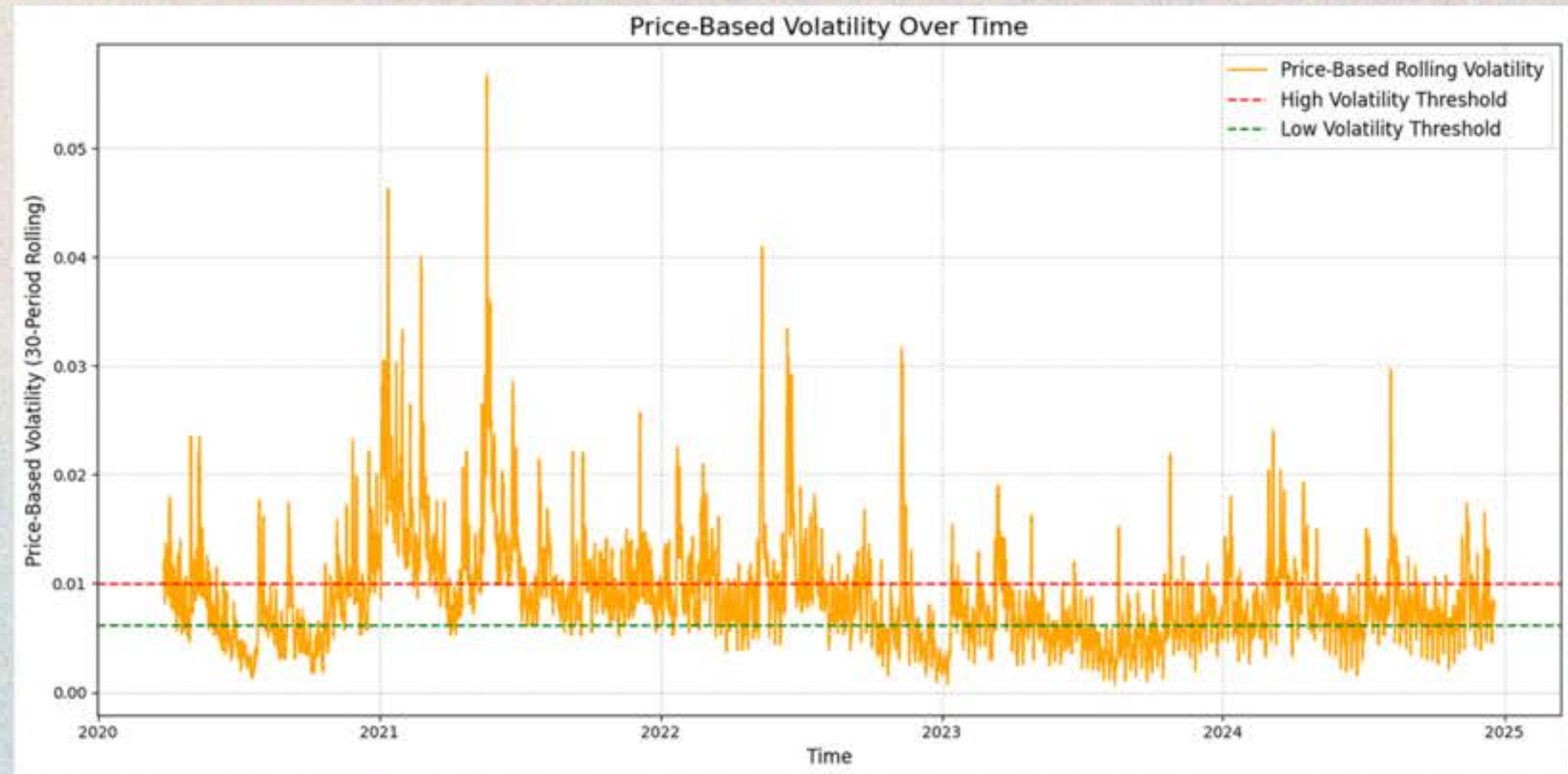
- **taker_sell_ratio:** The **strongest negative impact** on price_change (**Beta = -0.438, $p < 0.05$**). A higher taker sell ratio reflects **increased sell-side market activity**, aligning with bearish market sentiment.
- **coinbase_premium_gap:** A **significant positive impact** on price_change (**Beta = 0.129, $p < 0.05$**). A higher premium on Coinbase indicates **stronger demand**, positively affecting price.
- **long_liquidations:** A significant negative impact on price_change (**Beta = -0.1480, $p < 0.05$**). More long liquidations (**forced selling of long positions**) lead to price drops.

- **Non-Significant Variables:** Variables with a **p-value greater than 0.05** are considered statistically insignificant, indicating they do not have a reliable impact on price_change.
- From the result, **addresses_count_active** and **transactions_count_inflow** (**$p > 0.05$**) fall into this category and should not be prioritized for interpreting price changes.

2. How do these factors behave differently during periods of high versus low market volatility?

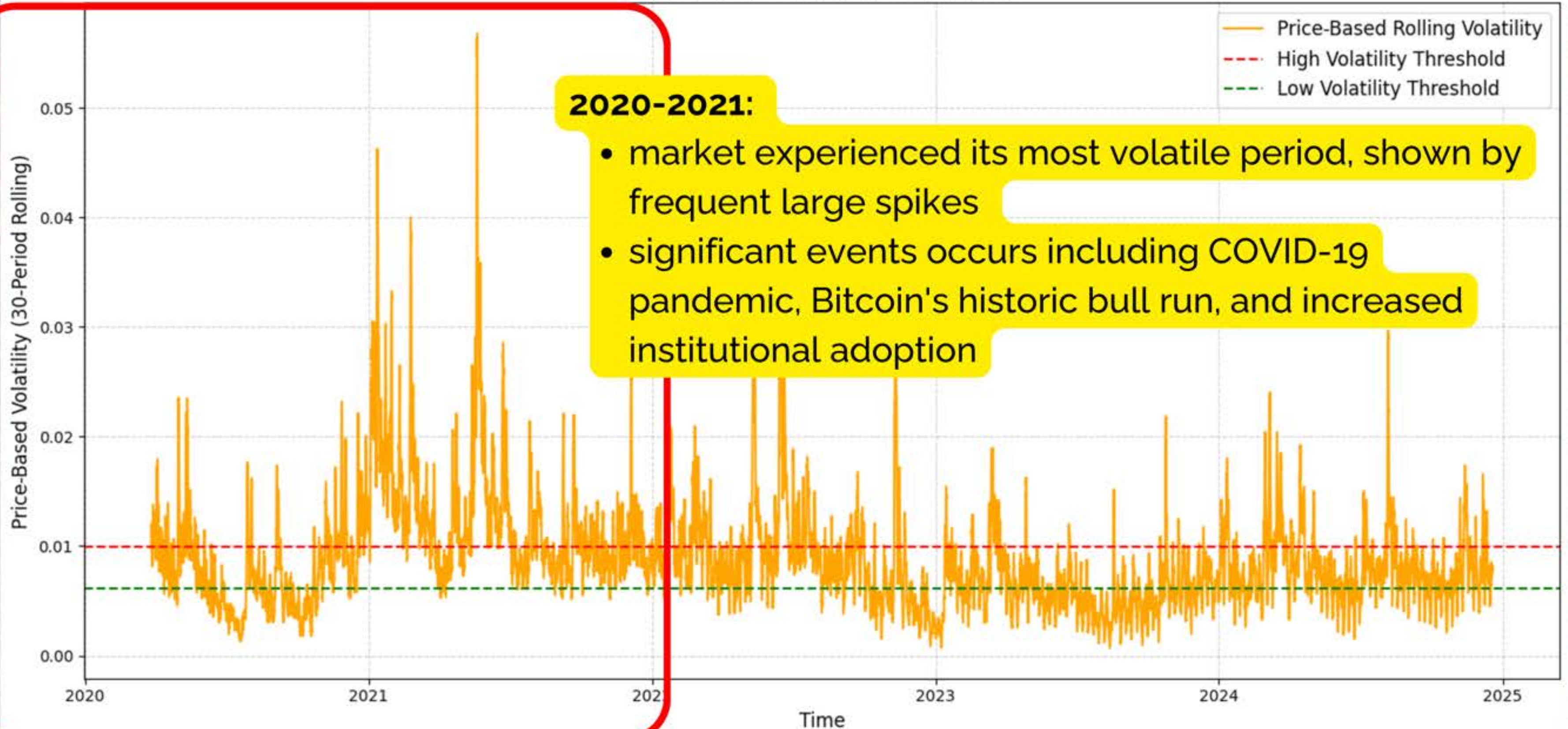
Methodology:

Based on the identified factors, we refined our analysis to focus on the 11 most significant factors. We first calculate the **price volatility** as a ratio of high to low prices, smoothed with a **30-day rolling average**, and classified into high and low categories using percentile thresholds. Next, we use **Random Forest and XGBoost models** to predict price volatility and their performance evaluated using MSE and R^2 scores. We then use **feature importance and correlation heatmaps** to reveal how factors behave differently during high and low volatility periods.



- High Volatility Threshold: 0.009912592109670222
- Low Volatility Threshold: 0.0061714496943869315
- progression from high to moderate volatility reflects the cryptocurrency market's transformation from a highly volatile to a more established market

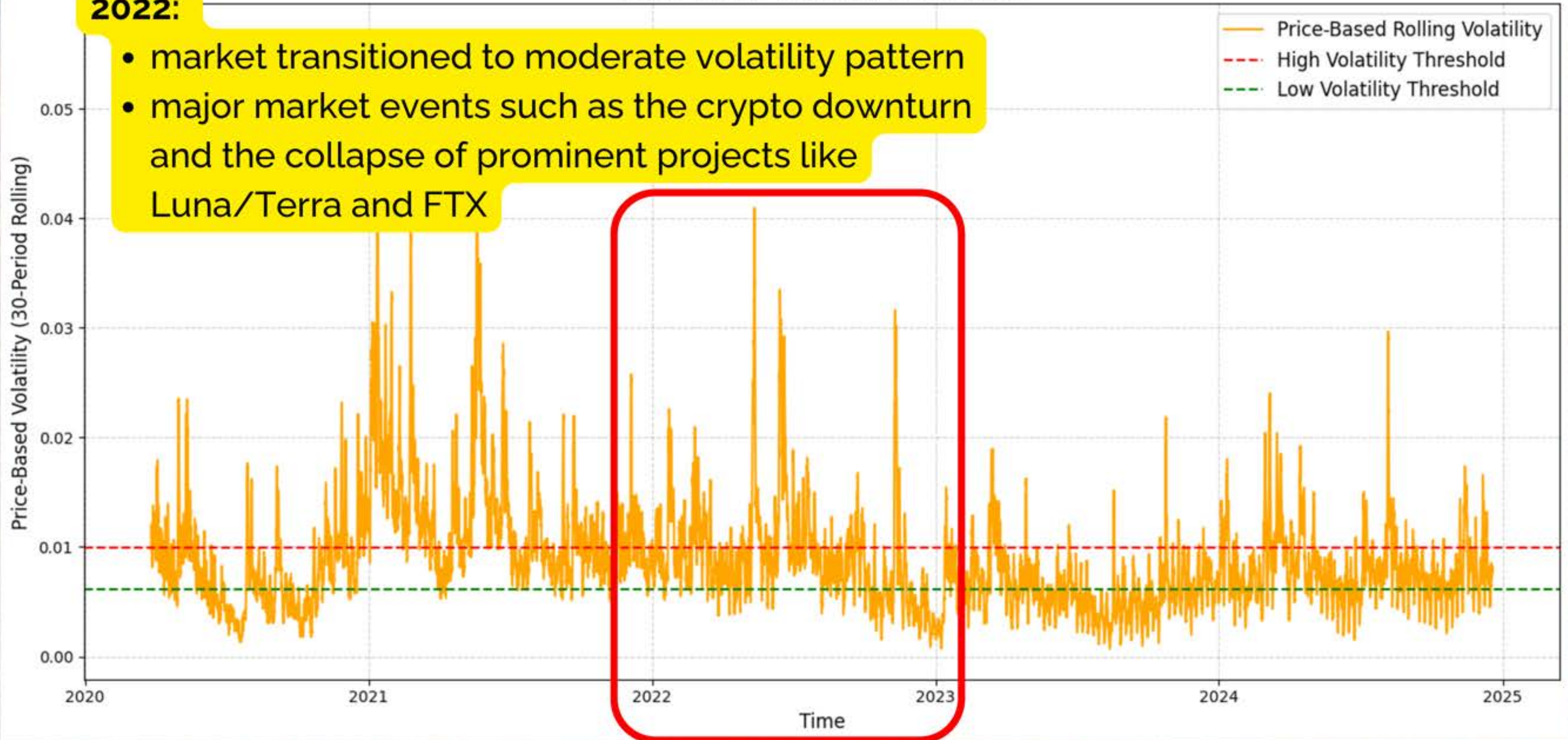
Price-Based Volatility Over Time



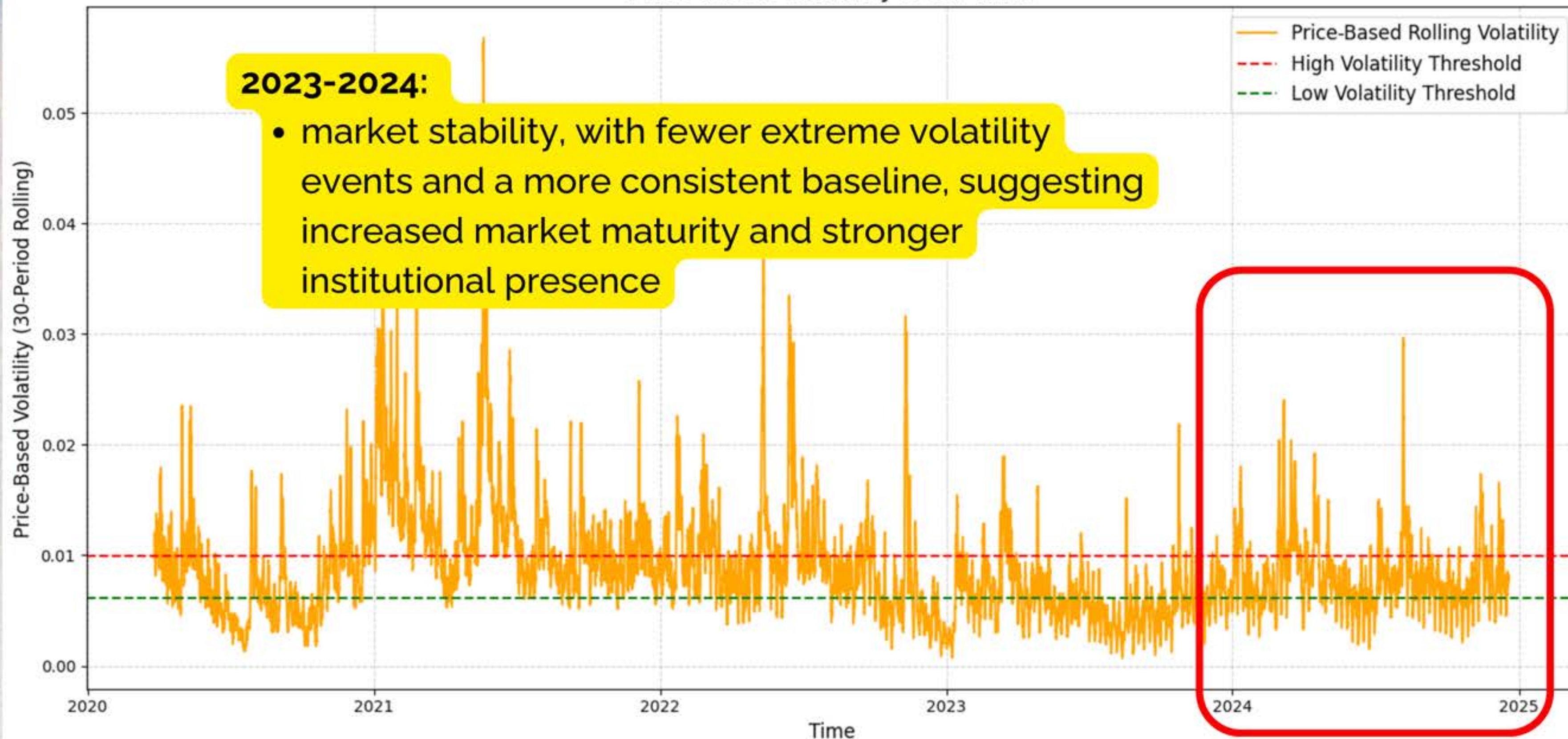
Price-Based Volatility Over Time

2022:

- market transitioned to moderate volatility pattern
- major market events such as the crypto downturn and the collapse of prominent projects like Luna/Terra and FTX



Price-Based Volatility Over Time



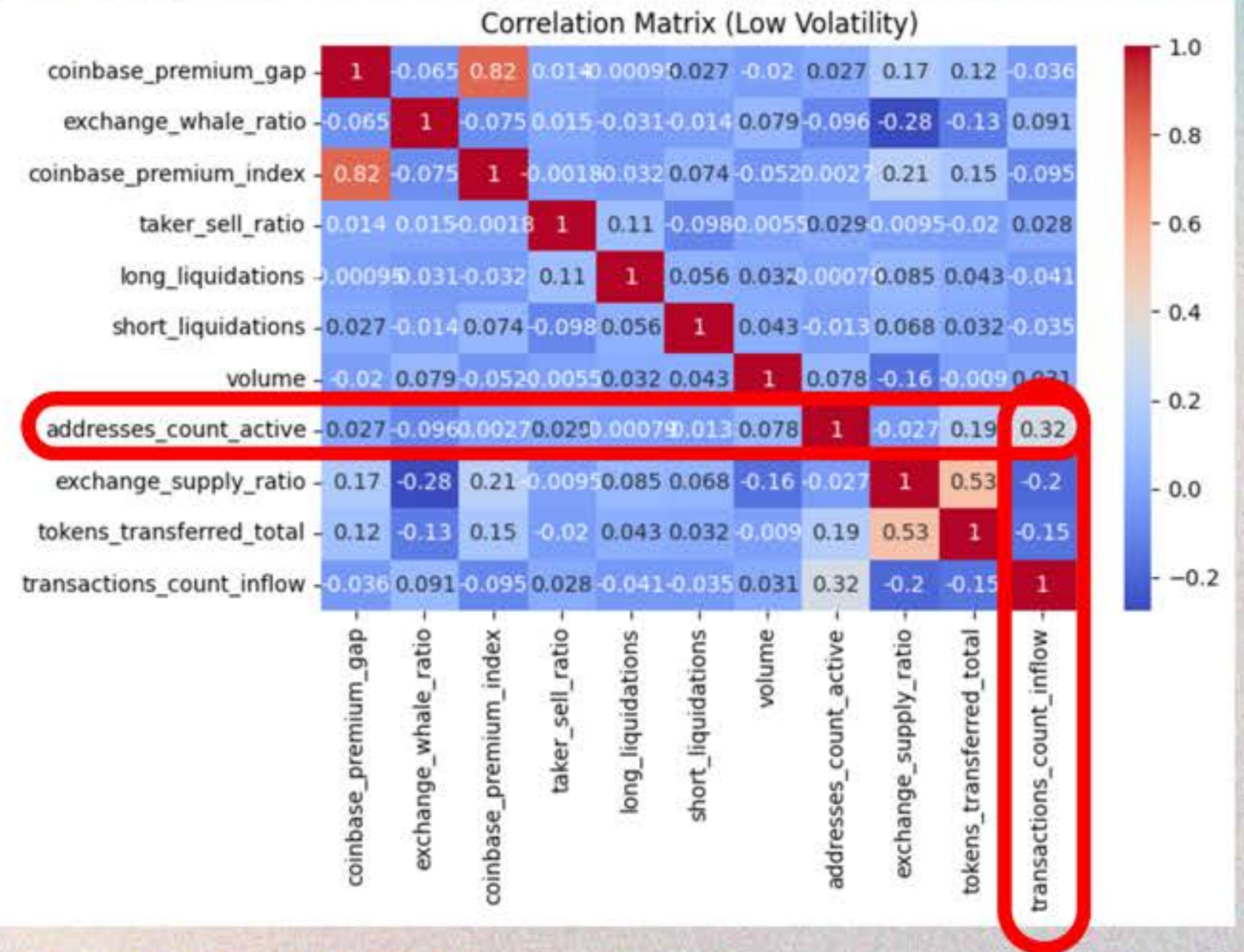
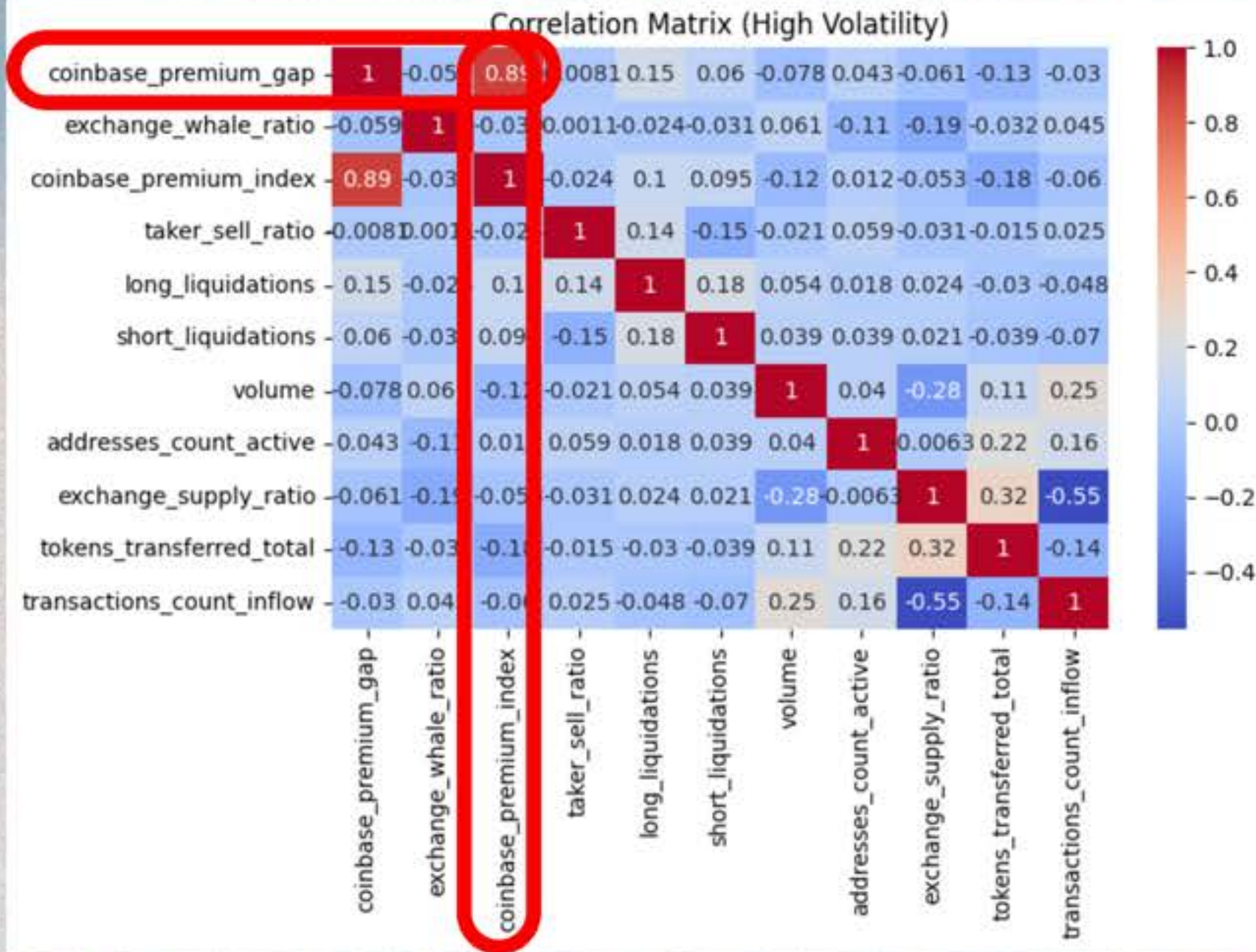
	Model	MSE	R2 Score
0	Random Forest	0.000016	0.777727
1	XGBoost	0.000014	0.808944

XGBoost outperformed Random Forest, achieving a slightly better mean squared error (MSE) and a higher R^2 score.

	Feature	Random Forest Importance	XGBoost Importance
4	long_liquidations	0.315861	0.322232
5	short_liquidations	0.228555	0.277645
6	volume	0.231176	0.151118
0	coinbase_premium_gap	0.048070	0.054083
8	exchange_supply_ratio	0.060073	0.047988
10	transactions_count_inflow	0.020075	0.047939
3	taker_sell_ratio	0.033223	0.032268
9	tokens_transferred_total	0.016119	0.021316
2	coinbase_premium_index	0.019801	0.020067
7	addresses_count_active	0.012400	0.013172
1	exchange_whale_ratio	0.014646	0.012172

Based on the ranking of the feature importance,

- **Liquidation Metrics (long_liquidations, short_liquidations)** is highly influential due to their direct relationship with market stress during high volatility.
- **Volume** is moderately important across all conditions, with slightly more predictive value in high volatility due to random spikes in trade.
- **Structural Metrics (exchange_supply_ratio, coinbase_premium_gap)** contribute to predictive power in low volatility as they capture more stable, fundamental market behaviors.



During High Volatility:

- Market activity is **more unpredictable** and relationships between factors tend to fluctuate.
- Key factors like *coinbase_premium_gap* and *coinbase_premium_index* align strongly, reflecting heightened investor sentiment and potential arbitrage.
- Liquidations become more relevant, reflecting stress in the market.

During Low Volatility:

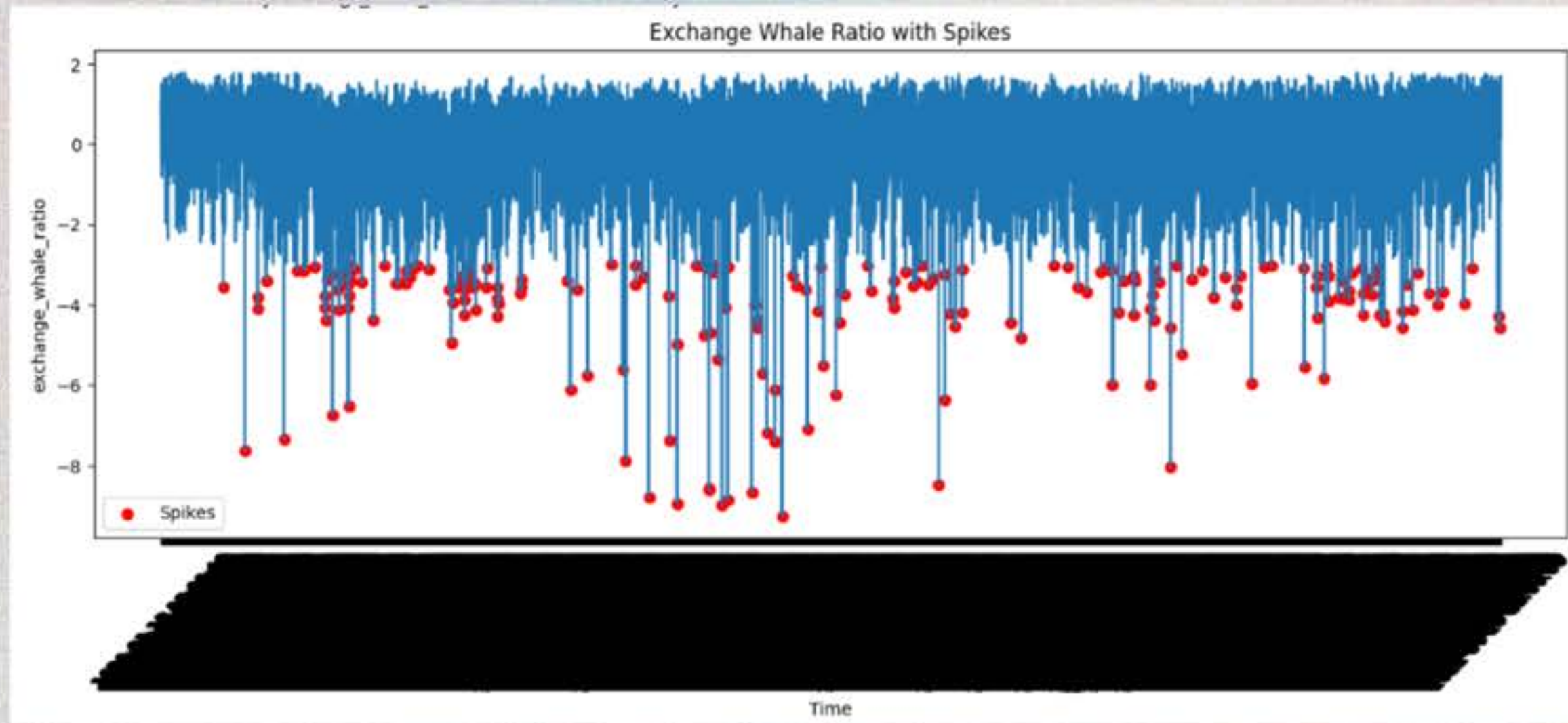
- Market behavior is **steadier**, and correlations between activity-based factors like *addresses_count_active* and *transactions_count_inflow* strengthen.
- **Exchange supply** becomes a more reliable indicator of overall activity.
- **Liquidations are less impactful.**



FINDINGS

- Since *long_liquidations* and *short_liquidations* are highly influential during periods of high volatility, it is crucial to develop monitoring systems or alerts for **sudden liquidation spikes** to serve as early indicators of market stress or trend reversals.
- During **low volatility**, *exchange_supply_ratio* and *coinbase_premium_gap* provide reliable insights into market stability. **Long-term investors** can use these metrics to assess market fundamentals and make strategic entries or exits.
- The **predictive significance of volume** during high volatility suggests opportunities to optimize trading strategies during periods of increased trading activity.

3. Are there any indicators in the dataset that signal potential market manipulation, such as sudden spikes in exchange_whale_ratio?



Number of spikes detected: 213

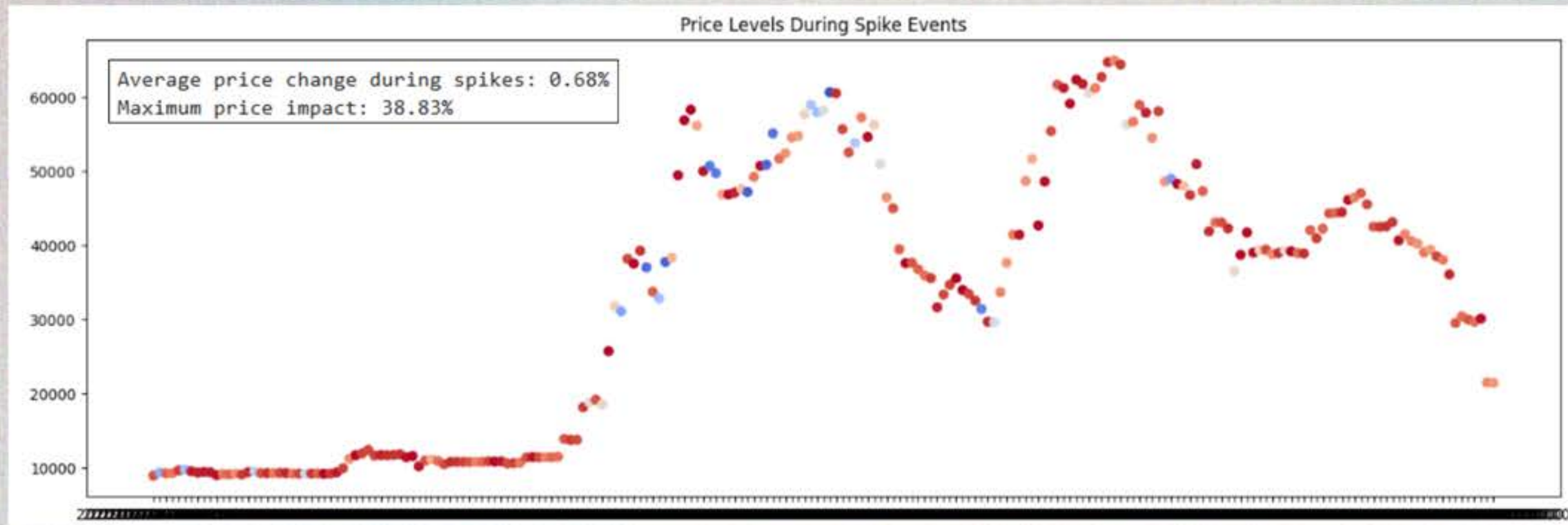
This metric represents the **ratio of large cryptocurrency holding (whales)** on exchanges where a sudden change in the ratio could indicate **coordinated buying or selling activity**, potentially aimed at manipulating the market.

Methodology:

The analysis uses statistical method, z-scores to identify spikes in the **exchange_whale_ratio** which measures how many standard deviations a data point is from the mean. A **high z-score** indicates an unusually large deviation which is considered a spike. These spikes may potentially become an indicators of market manipulation.

From this graph, we observed that

- blue line: exchange whale ratio
- red dots: spikes or unusual movements
- y-axis: values from -8 to 2
- most spikes occur between -4 to -8
- there are 213 detected spike events



This graph shows the cryptocurrency price levels specifically during spike events and indicates a clear correlation between **spike events** and **significant price movements**

- Price started around 10,000 and reached peaks near 60,000
- Average price change during spikes: 0.68%
- Maximum price impact: 38.14%
- Price changes significantly during spike events and their regular return to baseline after spikes indicates controlled movements

FINDINGS

- These statistics between the price movements during spikes suggest that spikes are a key driver of market volatility.
- There may be signs of **coordinated activity** because clusters of spikes and consistent large withdrawals suggest coordinated activity. Sudden trading volume surges without news may indicate that whales are trying to manipulate the market.
- For examples, whales might create an illusion of demand or supply to mislead traders, and leading to potential financial losses.
- Traders should be **cautious** during periods of unexplained trading volume surges or price spikes, as these may not reflect organic market conditions.



CONCLUSION & DISCUSSION

LESSONS LEARNT

High-quality, complete, and accurate data is essential for meaningful analysis. Visualizing data is critical for identifying trends and anomalies but requires careful design to avoid misinterpretation.

LIMITATIONS

Although statistical models can reveal patterns, but predicting future market behavior is uncertain due to the stochastic and speculative nature of cryptocurrency markets. Besides, external factors such as regulatory announcements, trends that are not captured in the dataset could significantly affect the market.

IMPLICATIONS OF STUDY

Traders can manage risk, or exploit potential trend reversals. Institutions can implement these systems to safeguard portfolios and identify early signs of market instability. Market analysts can enhance models predicting price trends and volatility.

THANK YOU!

<https://github.com/pang-lang/404Found>